

echap + :wq et pour écrire c'est i

pour relancer sur cmd :

ssh k8s@192.168.88.150

su -

kubectl get pods

kubectl get services

Job 1

Je commence par créer mes trois Vms sur lesquelles je vais travailler.

J'installe les outils de base :

```
Home x kubes-01 x
Réception de :7 http://deb.debian.org/debian bookworm/main amd64 vim-runtime all 2:9.0.1378-2+deb12u2 [7 027 kB]
Réception de :8 http://deb.debian.org/debian bookworm/main amd64 vim amd64 2:9.0.1378-2+deb12u2 [1 568 kB]
11,6 Mo réceptionnés en 1s (9 917 ko/s)
Sélection du paquet sudo précédemment désélectionné.
(Lecture de la base de données... 34192 fichiers et répertoires déjà installés.)
Préparation du dépaquetage de .../0-sudo_1.9.13p3-1+deb12u4_amd64.deb ...
Dépaquetage de sudo (1.9.13p3-1+deb12u4) ...
Sélection du paquet libcurl4:amd64 précédemment désélectionné.
Préparation du dépaquetage de .../1-libcurl4_7.88.1-10+deb12u14_amd64.deb ...
Dépaquetage de libcurl4:amd64 (7.88.1-10+deb12u14) ...
Sélection du paquet curl précédemment désélectionné.
Préparation du dépaquetage de .../2-curl_7.88.1-10+deb12u14_amd64.deb ...
Dépaquetage de curl (7.88.1-10+deb12u14) ...
Sélection du paquet libgpm2:amd64 précédemment désélectionné.
Préparation du dépaquetage de .../3-libgpm2_1.20.7-10+b1_amd64.deb ...
Dépaquetage de libgpm2:amd64 (1.20.7-10+b1) ...
Sélection du paquet libsodium23:amd64 précédemment désélectionné.
Préparation du dépaquetage de .../4-libsodium23_1.0.18-1+deb12u1_amd64.deb ...
Dépaquetage de libsodium23:amd64 (1.0.18-1+deb12u1) ...
Sélection du paquet net-tools précédemment désélectionné.
Préparation du dépaquetage de .../5-net-tools_2.10-0.1+deb12u2_amd64.deb ...
Dépaquetage de net-tools (2.10-0.1+deb12u2) ...
Sélection du paquet vim-runtime précédemment désélectionné.
Préparation du dépaquetage de .../6-vim-runtime_2%3a9.0.1378-2+deb12u2_all.deb ...
Ajout de « détournement de /usr/share/vim/vim90/doc/help.txt en /usr/share/vim/vim90/doc/help.txt.vim-tiny par vim-runtime »
Ajout de « détournement de /usr/share/vim/vim90/doc/tags en /usr/share/vim/vim90/doc/tags.vim-tiny par vim-runtime »
Dépaquetage de vim-runtime (2:9.0.1378-2+deb12u2) ...
Sélection du paquet vim précédemment désélectionné.
Préparation du dépaquetage de .../7-vim_2%3a9.0.1378-2+deb12u2_amd64.deb ...
Dépaquetage de vim (2:9.0.1378-2+deb12u2) ...
Paramétrage de net-tools (2.10-0.1+deb12u2) ...
Paramétrage de libsodium23:amd64 (1.0.18-1+deb12u1) ...
Paramétrage de libgpm2:amd64 (1.20.7-10+b1) ...
Paramétrage de sudo (1.9.13p3-1+deb12u4) ...
Paramétrage de libcurl4:amd64 (7.88.1-10+deb12u14) ...
Paramétrage de curl (7.88.1-10+deb12u14) ...
Paramétrage de vim-runtime (2:9.0.1378-2+deb12u2) ...
Paramétrage de vim (2:9.0.1378-2+deb12u2) ...
update-alternatives: utilisation de « /usr/bin/vim.basic » pour fournir « /usr/bin/ex » (ex) en mode automatique
update-alternatives: utilisation de « /usr/bin/vim.basic » pour fournir « /usr/bin/rview » (rview) en mode automatique
update-alternatives: utilisation de « /usr/bin/vim.basic » pour fournir « /usr/bin/rvim » (rvim) en mode automatique
update-alternatives: utilisation de « /usr/bin/vim.basic » pour fournir « /usr/bin/vi » (vi) en mode automatique
update-alternatives: utilisation de « /usr/bin/vim.basic » pour fournir « /usr/bin/view » (view) en mode automatique
update-alternatives: utilisation de « /usr/bin/vim.basic » pour fournir « /usr/bin/vim » (vim) en mode automatique
update-alternatives: utilisation de « /usr/bin/vim.basic » pour fournir « /usr/bin/vimdiff » (vimdiff) en mode automatique
Traitement des actions différées (« triggers ») pour man-db (2.11.2-2) ...
Traitement des actions différées (« triggers ») pour libc-bin (2.36-9+deb12u14) ...
root@kubes-01:~# apt install -y curl wget vim net-tools sudo
```

mon IP, lorsque je fais ip a, est : **192.168.88.150/24** brd 192.168.88.255

Je vais déclarer trois VMs à l'avance, donc éditer mon fichier pour fixer statiquement mes prochaines IPs. J'édite donc le fichier host.

```
Home x kubex-01 x
127.0.1.1      kubex-01.local  kubex-01

# The following lines are desirable for IPv6 capable hosts
::1          localhost ip6-localhost ip6-loopback
ff02::1      ip6-allnodes
ff02::2      ip6-allrouters
192.168.88.150 kubex-01.local kubex-01
192.168.88.151 kubex-02.local kubex-02
192.168.88.153 kubex-03.local kubex-03
~
~
```

Pour que ma VM kubex-01 garde toujours la même ip après un reboot, je vais changer quelques lignes :

```
Home x kubex-01 x

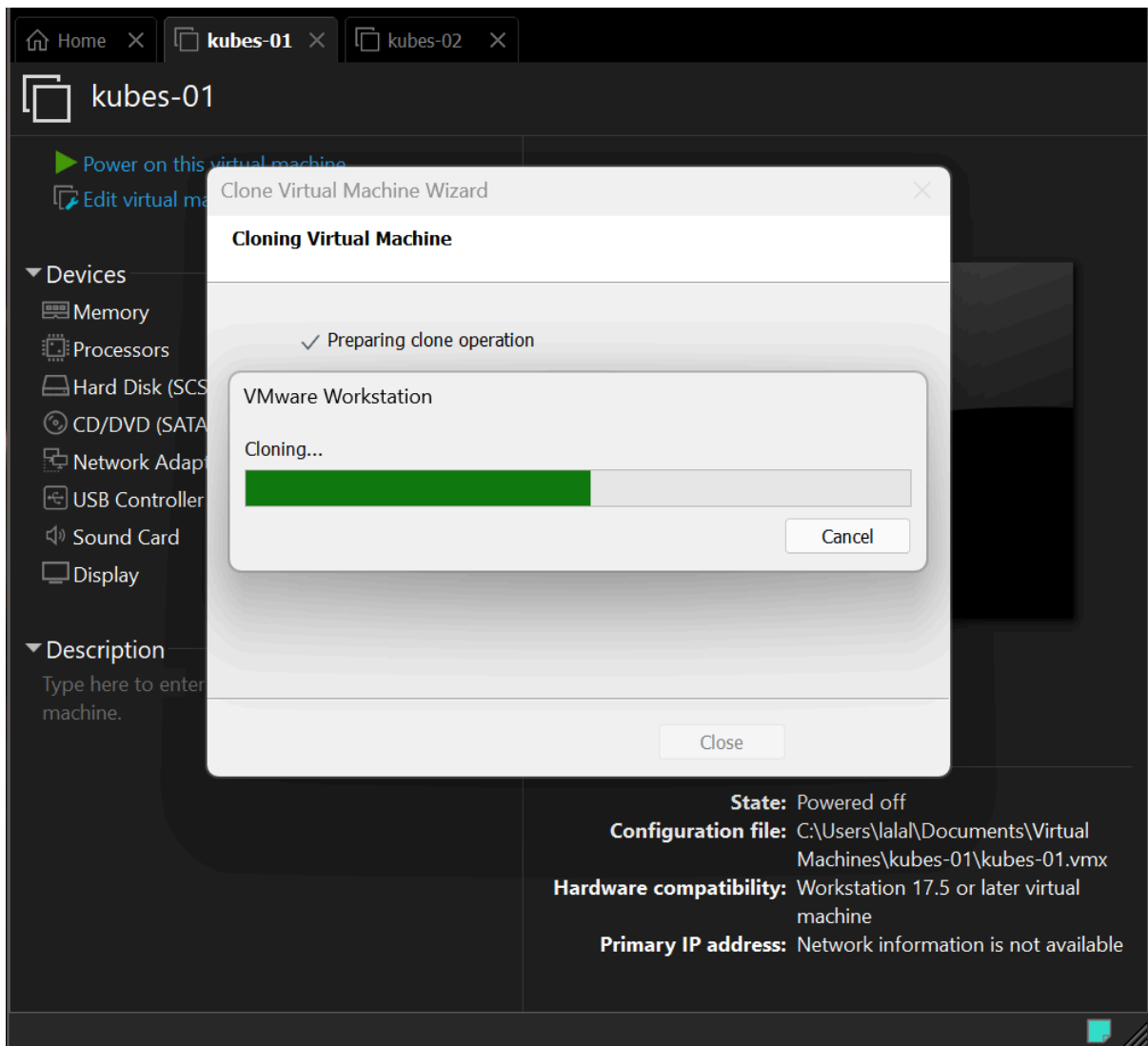
# This file describes the network interfaces available on your system
# and how to activate them. For more information, see interfaces(5).

source /etc/network/interfaces.d/*

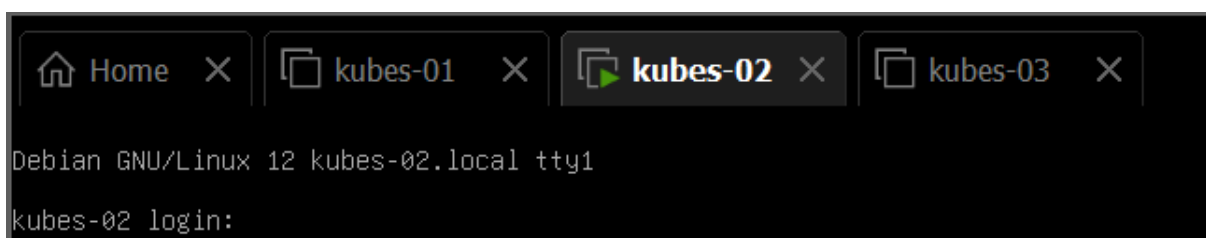
# The loopback network interface
auto lo
iface lo inet loopback

# The primary network interface
allow-hotplug ens33
iface ens33 inet static
    address 192.168.88.150
    netmask 255.255.255.0
    gateway 192.168.88.2
    dns-nameservers 8.8.8.8_
~
~
```

Je vais maintenant cloner ma première VM pour créer mes 3 VMs en tout.



Je change le nom et l'adresse IP de la deuxième VM, puis de la troisième.



```
Home x kubes-01 x kubes-02 x kubes-03 x
Debian GNU/Linux 12 kubes-01.local tty1

kubes-01 login: root
Password:
Linux kubes-01.local 6.1.0-49-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.1.174-1 (2026-05-26) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Mon Jun  8 11:08:53 CEST 2026 on tty1
root@kubes-01:~# hostnamectl set-hostname kubes-03.local
root@kubes-01:~# vim /etc/network/interfaces_
```

Je vais maintenant installer K3S sur mon kubes-01.

```
Home x kubes-01 x kubes-02 x kubes-03 x
[INFO] Using v1.35.5+k3s1 as release
[INFO] Downloading hash https://github.com/k3s-io/k3s/releases/download/v1.35.5%2Bk3s1/sha256sum-amd64.txt
[INFO] Downloading binary https://github.com/k3s-io/k3s/releases/download/v1.35.5%2Bk3s1/k3s
[INFO] Verifying binary download
[INFO] Installing k3s to /usr/local/bin/k3s
[INFO] Skipping installation of SELinux RPM
[INFO] Creating /usr/local/bin/kubectrl symlink to k3s
[INFO] Creating /usr/local/bin/crictl symlink to k3s
[INFO] Creating /usr/local/bin/ctr symlink to k3s
[INFO] Creating killall script /usr/local/bin/k3s-killall.sh
[INFO] Creating uninstall script /usr/local/bin/k3s-uninstall.sh
[INFO] env: Creating environment file /etc/systemd/system/k3s.service.env
[INFO] systemd: Creating service file /etc/systemd/system/k3s.service
[INFO] systemd: Enabling k3s unit
Created symlink /etc/systemd/system/multi-user.target.wants/k3s.service -> /etc/systemd/system/k3s.service.
[INFO] Host iptables-save/iptables-restore tools not found
[INFO] Host iptables-save/iptables-restore tools not found
[INFO] systemd: Starting k3s
root@kubes-01:~# [ 118.639522] tmpfs: Unknown parameter 'noswap'
systemctl status k3s
● k3s.service - Lightweight Kubernetes
   Loaded: loaded (/etc/systemd/system/k3s.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-06-08 11:49:47 CEST; 6min ago
     Docs: https://k3s.io
   Process: 945 ExecStartPre=/sbin/modprobe br_netfilter (code=exited, status=0/SUCCESS)
   Process: 948 ExecStartPre=/sbin/modprobe overlay (code=exited, status=0/SUCCESS)
   Main PID: 949 (k3s-server)
     Tasks: 81
    Memory: 1.3G
       CPU: 1min 28.572s
   CGroup: /system.slice/k3s.service
           └─ 949 "/usr/local/bin/k3s server"
              1009 "containerd"
              1752 /var/lib/rancher/k3s/data/921c0d40b0b3f73f2beae03ec1b23ec9aed3aa19fd651db6353e38f9636ddfe0/bin/containerd-s
              1846 /var/lib/rancher/k3s/data/921c0d40b0b3f73f2beae03ec1b23ec9aed3aa19fd651db6353e38f9636ddfe0/bin/containerd-s
              1946 /var/lib/rancher/k3s/data/921c0d40b0b3f73f2beae03ec1b23ec9aed3aa19fd651db6353e38f9636ddfe0/bin/containerd-s
              2874 /var/lib/rancher/k3s/data/921c0d40b0b3f73f2beae03ec1b23ec9aed3aa19fd651db6353e38f9636ddfe0/bin/containerd-s
              2977 /var/lib/rancher/k3s/data/921c0d40b0b3f73f2beae03ec1b23ec9aed3aa19fd651db6353e38f9636ddfe0/bin/containerd-s

Jun 08 11:50:52 kubes-01.local k3s[949]: I0608 11:50:52.839016 949 resource_quota_monitor.go:229] "QuotaMonitor created ob
Jun 08 11:50:52 kubes-01.local k3s[949]: I0608 11:50:52.839031 949 resource_quota_monitor.go:229] "QuotaMonitor created ob
Jun 08 11:50:52 kubes-01.local k3s[949]: I0608 11:50:52.839040 949 resource_quota_monitor.go:229] "QuotaMonitor created ob
Jun 08 11:50:52 kubes-01.local k3s[949]: I0608 11:50:52.839064 949 resource_quota_monitor.go:229] "QuotaMonitor created ob
Jun 08 11:50:52 kubes-01.local k3s[949]: I0608 11:50:52.839076 949 resource_quota_monitor.go:229] "QuotaMonitor created ob
Jun 08 11:50:52 kubes-01.local k3s[949]: I0608 11:50:52.839095 949 resource_quota_monitor.go:229] "QuotaMonitor created ob
Jun 08 11:50:52 kubes-01.local k3s[949]: I0608 11:50:52.839345 949 shared_informer.go:370] "Waiting for caches to sync"
Jun 08 11:50:53 kubes-01.local k3s[949]: I0608 11:50:53.052417 949 shared_informer.go:370] "Waiting for caches to sync"
Jun 08 11:50:54 kubes-01.local k3s[949]: I0608 11:50:54.140552 949 shared_informer.go:377] "Caches are synced"
Jun 08 11:50:54 kubes-01.local k3s[949]: I0608 11:50:54.155692 949 shared_informer.go:377] "Caches are synced"
lines 1-29/29 (END)
```

On voit bien que le texte en vert écrit “running”, et que tout fonctionne correctement.
Je vais récupérer le token pour pouvoir connecter les workers.

```
root@kubes-01:~# cat /var/lib/rancher/k3s/server/node-token
k10dd5e837bc2571aeaaab1167bcd1b21660866f04fbeb4d6b16bac07416921bd6::server:ab9799f58c57799e06a3e550dfd4764b
root@kubes-01:~#
```

Puis, j'active k3s sur mon kubens-02 :

```
[INFO] env: Creating environment file /etc/systemd/system/k3s.service.env
[INFO] systemd: Creating service file /etc/systemd/system/k3s.service
[INFO] systemd: Enabling k3s unit
Created symlink /etc/systemd/system/multi-user.target.wants/k3s.service → /etc/systemd/system/k3s.service.
[INFO] Host iptables-save/iptables-restore tools not found
[INFO] Host ip6tables-save/ip6tables-restore tools not found
[INFO] systemd: Starting k3s
root@kubes-02:~# [ 729.211812] tmpfs: Unknown parameter 'noswap'
systemctl status k3s
● k3s.service - Lightweight Kubernetes
   Loaded: loaded (/etc/systemd/system/k3s.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-06-08 12:27:28 CEST; 1min 42s ago
     Docs: https://k3s.io
   Process: 1582 ExecStartPre=/sbin/modprobe br_netfilter (code=exited, status=0/SUCCESS)
   Process: 1585 ExecStartPre=/sbin/modprobe overlay (code=exited, status=0/SUCCESS)
   Main PID: 1586 (k3s-server)
     Tasks: 82
    Memory: 1.4G
       CPU: 1min 15.135s
   CGroup: /system.slice/k3s.service
           └─1586 /usr/local/bin/k3s server"
           └─1646 "containerd "
```

```

[INFO] Using v1.35.5+k3s1 as release
[INFO] Downloading hash https://github.com/k3s-io/k3s/releases/download/v1.35.5%2Bk3s1/sha256sum-amd64.t
[INFO] Downloading binary https://github.com/k3s-io/k3s/releases/download/v1.35.5%2Bk3s1/k3s
[INFO] Verifying binary download
[INFO] Installing k3s to /usr/local/bin/k3s
[INFO] Skipping installation of SELinux RPM
[INFO] Creating /usr/local/bin/kubectrl symlink to k3s
[INFO] Creating /usr/local/bin/crictl symlink to k3s
[INFO] Creating /usr/local/bin/ctr symlink to k3s
[INFO] Creating killall script /usr/local/bin/k3s-killall.sh
[INFO] Creating uninstall script /usr/local/bin/k3s-uninstall.sh
[INFO] env: Creating environment file /etc/systemd/system/k3s.service.env
[INFO] systemd: Creating service file /etc/systemd/system/k3s.service
[INFO] systemd: Enabling k3s unit
Created symlink /etc/systemd/system/multi-user.target.wants/k3s.service → /etc/systemd/system/k3s.service
[INFO] Host iptables-save/iptables-restore tools not found
[INFO] Host ip6tables-save/ip6tables-restore tools not found
[INFO] systemd: Starting k3s
root@kubes-01:~# [ 2490.193606] tmpfs: Unknown parameter 'noswap'
systemctl status k3s
● k3s.service - Lightweight Kubernetes
   Loaded: loaded (/etc/systemd/system/k3s.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-06-08 13:05:48 CEST; 1min 23s ago
     Docs: https://k3s.io
  Process: 1370 ExecStartPre=/sbin/modprobe br_netfilter (code=exited, status=0/SUCCESS)
  Process: 1373 ExecStartPre=/sbin/modprobe overlay (code=exited, status=0/SUCCESS)

```

Je vais maintenant déployer les applications via K3S avec des fichiers YAML. Je parle d'nginx, apache, mariadb...

```
Home x kubex-01 x kubex-02 x kubex-03 x
kind: Deployment
metadata:
  name: nginx
spec:
  replicas: 1
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
        - name: nginx
          image: nginx:latest
          ports:
            - containerPort: 80
---
apiVersion: v1
kind: Service
metadata:
  name: nginx-service
spec:
  selector:
    app: nginx
  ports:
    - port: 80
      targetPort: 80
      nodePort: 30080
  type: NodePort
~
~
```

```
root@kubes-01:~/deployments# kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-5d47b8d59d-q8k5c             1/1     Running   0           7m57s
root@kubes-01:~/deployments# |
```

On peut voir que nginx running et que tout marche correctement !

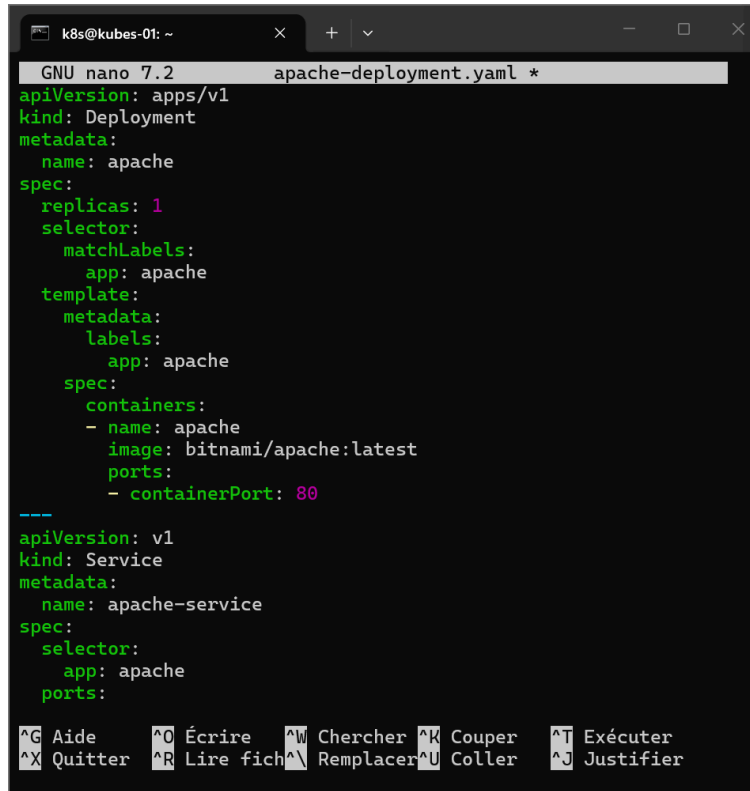
```
root@kubes-01:~/deployments# curl http://192.168.88.150:30080
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully in
stalled and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
root@kubes-01:~/deployments#
```

Hop lé, maintenant je vais faire Apache.

Lorsque j'ai voulu installer nginx, j'ai rencontré un problème d'images. Je vais donc remplacer dans mon deployment quelques lignes.



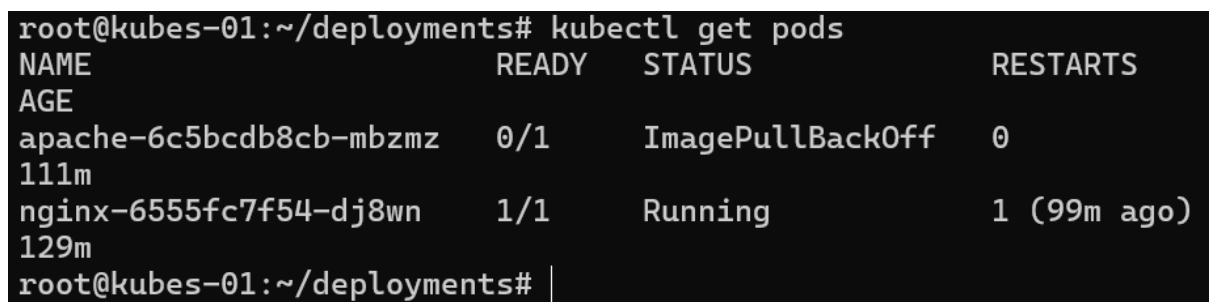
```
GNU nano 7.2 apache-deployment.yaml *
apiVersion: apps/v1
kind: Deployment
metadata:
  name: apache
spec:
  replicas: 1
  selector:
    matchLabels:
      app: apache
  template:
    metadata:
      labels:
        app: apache
    spec:
      containers:
      - name: apache
        image: bitnami/apache:latest
        ports:
        - containerPort: 80
---
apiVersion: v1
kind: Service
metadata:
  name: apache-service
spec:
  selector:
    app: apache
  ports:
```

Puis, comme ça ne marchait pas, j'ai remplacé par :

image: scr.io/linuxserver/apache:latest

En voyant que ça ne fonctionnait pas et qu'apache ne marchait pas, j'ai décidé de me créer un compte directement sur docker.com, en utilisant mon adresse mail La Plateforme_.

Voici le message qui montre bien qu'il n'y a aucune réponse sur apache :



```
root@kubes-01:~/deployments# kubectl get pods
NAME                                READY   STATUS              RESTARTS
AGE
apache-6c5bcd8cb-mbzmz              0/1     ImagePullBackOff    0
111m
nginx-6555fc7f54-dj8wn              1/1     Running              1 (99m ago)
129m
root@kubes-01:~/deployments# |
```

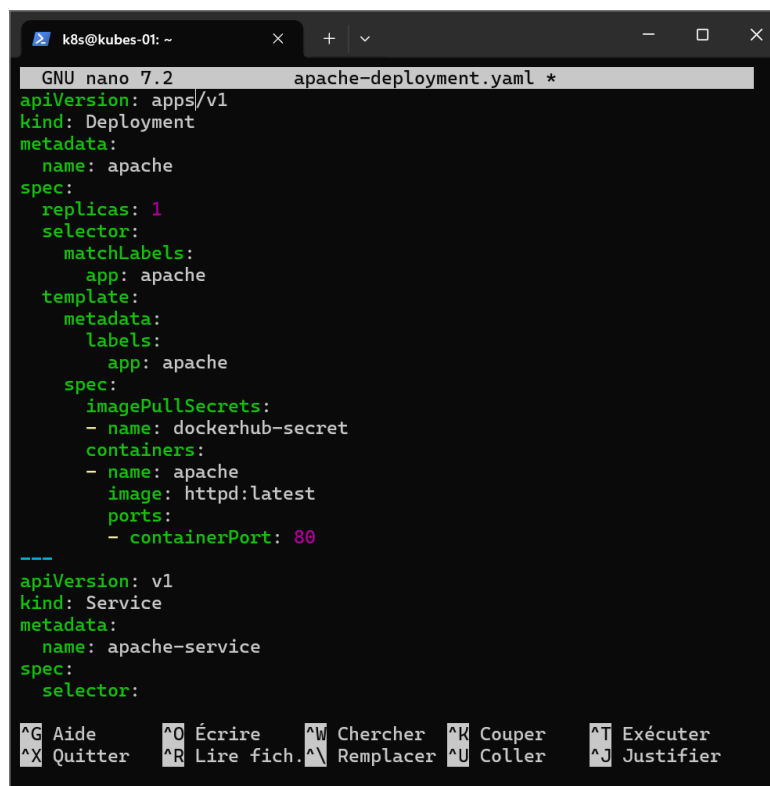
J'ai donc supprimé les anciennes données, et j'ai lié ma VM directement avec mes identifiants et mon mot de passe avec le compte que je venais de créer :

```

root@kubes-01:~/deployments# kubectl create secret docker-registry
dockerhub-secret \
  --docker-server=https://index.docker.io/v1/ \
  --docker-username=blookie \
  --docker-password=Alix2008*
secret/dockerhub-secret created
root@kubes-01:~/deployments# kubectl get secrets
NAME                                TYPE                                DATA  AGE
dockerhub-secret                    kubernetes.io/dockerconfigjson    1      7s
root@kubes-01:~/deployments#

```

Et là, on peut voir que le secret est bien créé. Je remplace donc dans le déploiement apache, l'image par mon dockerhub-secret :



```

GNU nano 7.2 apache-deployment.yaml *
apiVersion: apps/v1
kind: Deployment
metadata:
  name: apache
spec:
  replicas: 1
  selector:
    matchLabels:
      app: apache
  template:
    metadata:
      labels:
        app: apache
    spec:
      imagePullSecrets:
        - name: dockerhub-secret
      containers:
        - name: apache
          image: httpd:latest
          ports:
            - containerPort: 80
---
apiVersion: v1
kind: Service
metadata:
  name: apache-service
spec:
  selector:

```

Et... Apache marche enfin !

```

root@kubes-01:~/deployments# kubectl get pods
NAME                                READY  STATUS   RESTARTS  AGE
apache-5ffcd65847-fl6g5             1/1    Running  0          2m30s
nginx-6555fc7f54-dj8wn              1/1    Running  1 (112m ago)  142m
root@kubes-01:~/deployments# curl http://192.168.88.150:30081
<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01//EN" "http://www.w3.org/TR/html4/strict.dtd">
<html>
<head>
<title>It works! Apache httpd</title>
</head>
<body>
<p>It works!</p>
</body>
</html>
root@kubes-01:~/deployments#

```


Je vais maintenant passer à l'installation de MariaDB.

```
GNU nano 7.2 mariadb-deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: mariadb
spec:
  replicas: 1
  selector:
    matchLabels:
      app: mariadb
  template:
    metadata:
      labels:
        app: mariadb
    spec:
      imagePullSecrets:
        - name: dockerhub-secret
      containers:
        - name: mariadb
          image: mariadb:latest
          ports:
            - containerPort: 3306
          env:
            - name: MYSQL_ROOT_PASSWORD
              value: "rootpassword"
            - name: MYSQL_DATABASE
              value: "mydb"
---
apiVersion: v1
```

[Lecture de 39 lignes]

^G Aide ^O Écrire ^W Chercher ^K Couper ^T Exécuter
^X Quitter ^R Lire fich. ^\ Remplacer ^U Coller ^J Justifier

Et MariaDB fonctionne parfaitement aussi !

```
root@kubes-01:~/deployments# kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
apache-5ffcd65847-fl6g5             1/1     Running   0           7m41s
mariadb-5cdd94f94d-rtmwc            1/1     Running   0           2m35s
nginx-6555fc7f54-dj8wn              1/1     Running   1 (118m ago) 147m
root@kubes-01:~/deployments# curl http://192.168.88.150:30306
curl: (1) Received HTTP/0.9 when not allowed
```

Il n'y a pas d'autre réponse parce que MariaDB n'est pas un serveur HTTP, il parle le protocole MySQL.

Maintenant que j'ai téléchargé toutes mes applications sur le kubes-01, je vais copier tout ça sur le kubes-02 et le kubes-03.

```
Windows PowerShell
Copyright (C) Microsoft Corporation. Tous droits réservés.

Installez la dernière version de PowerShell pour de nouvelles fonctionnalités et améliorations ! https://aka.ms/PSWindows

PS C:\Users\lalal> scp k8s@192.168.88.150:/home/k8s/*.yaml k8s@192.168.88.151:/home/k8s/
k8s@192.168.88.151's password:
k8s@192.168.88.150's password:
apache-deployment.yaml              100% 542   132.3KB/s   00:00
mariadb-deployment.yaml             100% 692   225.3KB/s   00:00
nginx-deployment.yaml               100% 515   167.6KB/s   00:00
PS C:\Users\lalal>
```

Et pareil avec kubes-03 :

```
k8s@kubes-02:~$ scp k8s@192.168.88.150:/home/k8s/*.yaml k8s@192.168.88.153:/home/k8s/
The authenticity of host '192.168.88.153 (192.168.88.153)' can't be established.
ED25519 key fingerprint is SHA256:FJ/dySz0bw458xjD26IltXPSBZUQFyqZe1A+x5sV868.
This host key is known by the following other names/addresses:
C:\Users\lala\ssh\known_hosts:3: 192.168.88.150
C:\Users\lala\ssh\known_hosts:6: 192.168.88.151
Are you sure you want to continue connecting (yes/no/[fingerprint])?
Warning: Permanently added '192.168.88.153' (ED25519) to the list of known hosts.
k8s@192.168.88.153's password:
k8s@192.168.88.150's password:
apache-deployment.yaml      100% 542   176.4KB/s   00:00
mariadb-deployment.yaml     100% 692   337.9KB/s   00:00
nginx-deployment.yaml       100% 515   251.5KB/s   00:00
PS C:\Users\lala>
```

Je vais créer le secret Docker sur mon kubes-02, comme fait précédemment avec mon kubes-01.

Voilà ce que je fais pour tout copier. A la fin, on peut bien remarquer que c'est en train de se créer. Il faut donc attendre.

```
k8s@kubes-02:~$ su
Mot de passe :
root@kubes-02:/home/k8s# kubectl create secret docker-registry dock
erhub-secret \
  --docker-server=https://index.docker.io/v1/ \
  --docker-username=blookie \
  --docker-password=Alix2008*
secret/dockerhub-secret created
root@kubes-02:/home/k8s# mv /home/k8s/*.yaml /root/
root@kubes-02:/home/k8s# cd /root
root@kubes-02:~# kubectl apply -f nginx-deployment.yaml
deployment.apps/nginx created
service/nginx-service created
root@kubes-02:~# kubectl apply -f apache-deployment.yaml
deployment.apps/apache created
service/apache-service created
root@kubes-02:~# kubectl apply -f mariadb-deployment.yaml
deployment.apps/mariadb created
service/mariadb-service created
root@kubes-02:~# kubectl get pods
NAME                                READY    STATUS              RESTARTS   A
GE
apache-5ffcd65847-k6m8v            0/1      ContainerCreating   0           1
ls
mariadb-5cdd94f94d-8w2qq           0/1      ContainerCreating   0           6
s
nginx-6555fc7f54-g6p65             1/1      Running             0           1
7s
root@kubes-02:~#
```

Après une trentaine de secondes, voici les trois applications qui fonctionnent :

```
root@kubes-02:~# kubectl get pods
NAME                                READY    STATUS    RESTARTS   AGE
apache-5ffcd65847-k6m8v            1/1      Running   0           2m2s
mariadb-5cdd94f94d-8w2qq           1/1      Running   0           117s
nginx-6555fc7f54-g6p65             1/1      Running   0           2m8s
root@kubes-02:~#
```

Je vais donc réitérer pour kubes-03 ! C'est exactement les mêmes commandes que pour kubes-02, mais cette fois je me connecte à un nouvel onglet Powershell à ma VM.

```
root@kubes-03:/home/k8s# kubectl create secret docker-registry dock
erhub-secret \
> --docker-server=https://index.docker.io/v1/ \
--docker-username=blookie \
--docker-password=Alix2008*
secret/dockerhub-secret created
root@kubes-03:/home/k8s# mv /home/k8s/*.yaml /root/
root@kubes-03:/home/k8s# cd /root
root@kubes-03:~# kubectl apply -f nginx-deployment.yaml
deployment.apps/nginx created
service/nginx-service created
root@kubes-03:~# kubectl apply -f apache-deployment.yaml
deployment.apps/apache created
service/apache-service created
root@kubes-03:~# kubectl apply -f mariadb-deployment.yaml
deployment.apps/mariadb created
service/mariadb-service created
```

On attend que tout se mette en place... Et après avoir attendu :

```
root@kubes-03:~# kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
apache-5ffcd65847-z9m65             1/1     Running   0           82s
mariadb-5cdd94f94d-6gjdz            1/1     Running   0           75s
nginx-6555fc7f54-cjrw5              1/1     Running   0           87s
root@kubes-03:~#
```

Job 3

Maintenant, je vais récupérer le token du kubes-01. Le voici :

```
root@kubes-01:~/deployments# cat /var/lib/rancher/k3s/server/node-t
oken
K10dd5e837bc2571aeaeab1167bcd1b21660866f04fbeb4d6b16bac07416921bd6
::server:ab9799f58c57799e06a3e550dfd4764b
root@kubes-01:~/deployments#
```

Je le mets bien de côté :

```
K10dd5e837bc2571aeaeab1167bcd1b21660866f04fbeb4d6b16bac07416921bd6::server:a
b9799f58c57799e06a3e550dfd4764b
```

Puis, je me connecte à mon kubes-02 et à mon kubes-03.

```
k8s@kubes x k8s@kubes x k8s@kubes x + - □ x
t.service → /etc/systemd/system/k3s-agent.service.
[INFO] Host iptables-save/iptables-restore tools not found
[INFO] Host ip6tables-save/ip6tables-restore tools not found
[INFO] systemd: Starting k3s-agent
root@kubes-03:/home/k8s# systemctl start k3s-agent
root@kubes-03:/home/k8s# systemctl status k3s-agent
● k3s-agent.service - Lightweight Kubernetes
   Loaded: loaded (/etc/systemd/system/k3s-agent.service; enabled>
   Active: active (running) since Tue 2026-06-09 08:38:00 CEST; 1>
     Docs: https://k3s.io
   Process: 3811 ExecStartPre=/sbin/modprobe br_netfilter (code=ex>
   Process: 3812 ExecStartPre=/sbin/modprobe overlay (code=exited,>
   Main PID: 3813 (k3s-agent)
     Tasks: 43
    Memory: 97.2M
       CPU: 6.789s
   CGroup: /system.slice/k3s-agent.service
           └─3813 "/usr/local/bin/k3s agent"
             └─3859 "containerd "
               └─4135 /var/lib/rancher/k3s/data/921c0d40b0b3f73f2beae>

juin 09 08:38:02 kubes-03.local k3s[3813]: I0609 08:38:02.445801
juin 09 08:38:02 kubes-03.local k3s[3813]: I0609 08:38:02.453864
juin 09 08:38:02 kubes-03.local k3s[3813]: E0609 08:38:02.455147
juin 09 08:38:02 kubes-03.local k3s[3813]: I0609 08:38:02.455306
juin 09 08:38:02 kubes-03.local k3s[3813]: I0609 08:38:02.455361
juin 09 08:38:02 kubes-03.local k3s[3813]: E0609 08:38:02.455856
juin 09 08:38:02 kubes-03.local k3s[3813]: I0609 08:38:02.456448
juin 09 08:38:02 kubes-03.local k3s[3813]: I0609 08:38:02.456478
juin 09 08:38:02 kubes-03.local k3s[3813]: I0609 08:38:02.464697
juin 09 08:38:02 kubes-03.local k3s[3813]: I0609 08:38:02.472760
lines 1-25/25 (END)
```

On peut voir qu'il y a écrit "enabled" et "active (running)".

```
root@kubes-01:/home/k8s# kubectl get nodes
NAME                STATUS    ROLES    AGE     VERSION
kubes-01.local      Ready     control-plane 21h     v1.35.5+k3s1
kubes-02.local      Ready     <none>      13h     v1.35.5+k3s1
kubes-03.local      Ready     <none>      19m     v1.35.5+k3s1
root@kubes-01:/home/k8s#
```

Voilà mon Kuster complet, toutes les machines sont connectées à la première (kubes-01), et elle devient le Master.

```
root@kubes-01:/home/k8s# kubectl get pods -o wide
NAME                READY  STATUS    RESTARTS      AGE  IP
NODE                NOMINATED NODE  READINESS GATES
apache-5ffcd65847-fl6g5 1/1    Running   1 (62m ago)   15h  10.
42.0.29 kubes-01.local <none>      <none>
mariadb-5cdd94f94d-rtmwc 1/1    Running   1 (62m ago)   15h  10.
42.0.27 kubes-01.local <none>      <none>
nginx-6555fc7f54-dj8wn 1/1    Running   2 (62m ago)   17h  10.
42.0.28 kubes-01.local <none>      <none>
root@kubes-01:/home/k8s#
```

Comme on peut le voir, mes applications tournent bien. Par contre, on remarque que tous les pods sont encore en kubes-01, mais c'est normal puisque je les avais déployés avant de créer le kluster. Mais au prochain Job, je supprimerai tout et je re-déployerai avec des replicas, pour que les pods se répartissent sur les 3 nodes.

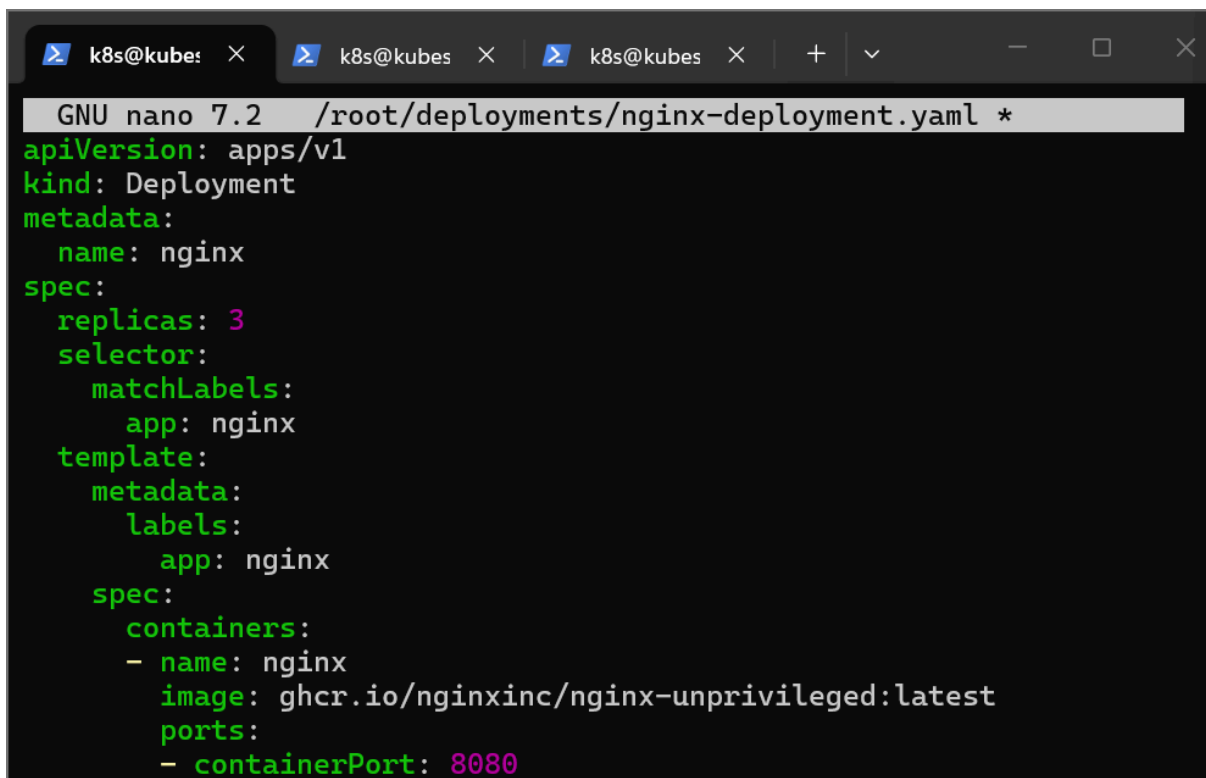
Job 4

Je vais maintenant supprimer toutes les applications. Dans mon kubes-01, je fais ces lignes de commande :

```
root@kubes-01:/home/k8s# cd /root/deployments
root@kubes-01:~/deployments# kubectl delete -f nginx-deployment.yaml
deployment.apps "nginx" deleted from default namespace
service "nginx-service" deleted from default namespace
root@kubes-01:~/deployments# kubectl delete -f apache-deployment.yaml
deployment.apps "apache" deleted from default namespace
service "apache-service" deleted from default namespace
root@kubes-01:~/deployments# kubectl delete -f mariadb-deployment.yaml
deployment.apps "mariadb" deleted from default namespace
service "mariadb-service" deleted from default namespace
root@kubes-01:~/deployments# kubectl get pods
No resources found in default namespace.
root@kubes-01:~/deployments#
```

Comme on peut le voir, il y a bien marqué "No resources found in default namespace." Maintenant, je vais tout réinstaller. Je vais modifier chaque fichier YAML pour mettre replicas:3 (un pod par note).

Je vais commencer par nginx.



```
k8s@kubes x k8s@kubes x k8s@kubes x + v - □ ×
GNU nano 7.2 /root/deployments/nginx-deployment.yaml *
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx
spec:
  replicas: 3
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
      - name: nginx
        image: ghcr.io/nginxinc/nginx-unprivileged:latest
        ports:
        - containerPort: 8080
```

Je change mon replicas: 1 à mon replicas: 3.

```

root@kubes-01:~/deployments# kubectl apply -f /root/deployments/nginx-deployment.yaml
deployment.apps/nginx created
service/nginx-service created
root@kubes-01:~/deployments# |

```

Puis je passe à Apache ! Je fais le même principe, en remplaçant les replicas de 1 par 3.

```

root@kubes-01:~/deployments# kubectl apply -f /root/deployments/apache-deployment.yaml
deployment.apps/apache created
service/apache-service created
root@kubes-01:~/deployments#

```

C'est le même principe ici pour MariaDB.

```

root@kubes-01:~/deployments# nano /root/deployments/mariadb-deployment.yaml
root@kubes-01:~/deployments# kubectl apply -f /root/deployments/mariadb-deployment.yaml
deployment.apps/mariadb created
service/mariadb-service created
root@kubes-01:~/deployments#

```

```

root@kubes-01:~/deployments# kubectl get pods -o wide
NAME                                READY   STATUS    RESTARTS   AGE   IP
NODE                                NOMINATED NODE   READINESS GATES
apache-5ffcd65847-6mcb5             1/1     Running   0          23m   10.42.1.4
kubes-02.local                      <none>   <none>
apache-5ffcd65847-g4bcv             1/1     Running   0          23m   10.42.2.3
kubes-03.local                      <none>   <none>
apache-5ffcd65847-htrbg             1/1     Running   0          23m   10.42.0.33
kubes-01.local                      <none>   <none>
mariadb-5cdd94f94d-dd9df            1/1     Running   0          69s   10.42.2.4
kubes-03.local                      <none>   <none>
mariadb-5cdd94f94d-lfcdx            1/1     Running   0          69s   10.42.0.34
kubes-01.local                      <none>   <none>
mariadb-5cdd94f94d-qtrcz            1/1     Running   0          69s   10.42.1.5
kubes-02.local                      <none>   <none>
nginx-6555fc7f54-4795l              1/1     Running   0          33m   10.42.2.2
kubes-03.local                      <none>   <none>
nginx-6555fc7f54-4q9zw              1/1     Running   0          33m   10.42.1.3
kubes-02.local                      <none>   <none>
nginx-6555fc7f54-8bzdq              1/1     Running   0          33m   10.42.0.32
kubes-01.local                      <none>   <none>
root@kubes-01:~/deployments#

```

On peut voir que les pods sont bien répartis sur les trois nodes.

Maintenant, je vais suspendre kubes-03, et voir s'il y a une redistribution des pods sur kubes-01 ou kubes-02.

```
root@kubes-01:~/deployments# kubectl get pods -o wide
NAME                                READY   STATUS    RESTARTS   AGE   I
P      NODE      NOMINATED NODE   READINESS GATES
apache-5ffcd65847-6mcb5             1/1    Running   0           75m   1
0.42.1.4    kubes-02.local    <none>      <none>
apache-5ffcd65847-8tz6c             1/1    Running   0           44m   1
0.42.1.6    kubes-02.local    <none>      <none>
apache-5ffcd65847-g4bcv             1/1    Terminating 0           75m   1
0.42.2.3    kubes-03.local    <none>      <none>
apache-5ffcd65847-htrbg             1/1    Running   0           75m   1
0.42.0.33   kubes-01.local    <none>      <none>
mariadb-5cdd94f94d-8xx9f           1/1    Running   0           44m   1
0.42.1.8    kubes-02.local    <none>      <none>
mariadb-5cdd94f94d-dd9df           1/1    Terminating 0           52m   1
0.42.2.4    kubes-03.local    <none>      <none>
mariadb-5cdd94f94d-lfcdx           1/1    Running   0           52m   1
0.42.0.34   kubes-01.local    <none>      <none>
mariadb-5cdd94f94d-qtrcz           1/1    Running   0           52m   1
0.42.1.5    kubes-02.local    <none>      <none>
nginx-6555fc7f54-4795l             1/1    Terminating 0           85m   1
0.42.2.2    kubes-03.local    <none>      <none>
nginx-6555fc7f54-4q9zw             1/1    Running   0           85m   1
0.42.1.3    kubes-02.local    <none>      <none>
nginx-6555fc7f54-8bzd             1/1    Running   0           85m   1
0.42.0.32   kubes-01.local    <none>      <none>
nginx-6555fc7f54-kw5x4             1/1    Running   0           44m   1
0.42.1.7    kubes-02.local    <none>      <none>
root@kubes-01:~/deployments#
```

On peut voir qu'il n'y a aucune interruption de service, les pods de kubes-03 sont en Terminating (en train de s'arrêter) et Kubernetes a automatiquement redéployé de nouveaux pods sur kubes-01 et kubes-02.

```

root@kubes-01:~/deployments# kubectl get pods -o wide
NAME                                READY   STATUS    RESTARTS   AGE     IP
NODE                                NOMINATED NODE   READINESS GATES
apache-5ffcd65847-6mcb5             1/1     Running   0           108m    10.4
2.1.4    kubes-02.local    <none>     <none>
apache-5ffcd65847-8tz6c             1/1     Running   0           77m     10.4
2.1.6    kubes-02.local    <none>     <none>
apache-5ffcd65847-htrbg             1/1     Running   0           108m    10.4
2.0.33   kubes-01.local    <none>     <none>
mariadb-5cdd94f94d-8xx9f           1/1     Running   0           77m     10.4
2.1.8    kubes-02.local    <none>     <none>
mariadb-5cdd94f94d-lfcdz           1/1     Running   0           85m     10.4
2.0.34   kubes-01.local    <none>     <none>
mariadb-5cdd94f94d-qtrcz           1/1     Running   0           85m     10.4
2.1.5    kubes-02.local    <none>     <none>
nginx-6555fc7f54-4q9zw             1/1     Running   0           118m    10.4
2.1.3    kubes-02.local    <none>     <none>
nginx-6555fc7f54-8bzdj             1/1     Running   0           118m    10.4
2.0.32   kubes-01.local    <none>     <none>
nginx-6555fc7f54-kw5x4             1/1     Running   0           77m     10.4
2.1.7    kubes-02.local    <none>     <none>
root@kubes-01:~/deployments# kubectl get nodes
NAME                STATUS    ROLES    AGE     VERSION
kubes-01.local      Ready     control-plane   24h     v1.35.5+k3s1
kubes-02.local      Ready     <none>         16h     v1.35.5+k3s1
kubes-03.local      Ready     <none>        3h30m    v1.35.5+k3s1
root@kubes-01:~/deployments#

```

Les 3 nodes sont Ready et tous les pods sont Running !

Job 5

Je vais créer un PersistentVolume sur Nginx


```
GNU nano 7.2 /root/deployments/nginx-pv.yaml
apiVersion: v1
kind: PersistentVolume
metadata:
  name: nginx-pv
spec:
  capacity:
    storage: 1Gi
  accessModes:
    - ReadWriteOnce
  hostPath:
    path: /data/nginx
---
apiVersion: v1
kind: PersistentVolumeClaim
metadata:
  name: nginx-pvc
spec:
  accessModes:
    - ReadWriteOnce
  resources:
    requests:
      storage: 1Gi
```

Le PV est créé, mais le PVC est en pending. C'est parce que k3s utilise local-path comme storage class par défaut. Je vais donc adapter le PV pour qu'il soit compatible.

```
GNU nano 7.2 /root/deployments/nginx-pv.yaml *
apiVersion: v1
kind: PersistentVolumeClaim
metadata:
  name: nginx-pvc
spec:
  accessModes:
    - ReadWriteOnce
  storageClassName: local-path
  resources:
    requests:
      storage: 1Gi
```

Mes pods nginx sont en Pending à cause du PVC. Je vais essayer de régler le problème.

```

root@kubes-01:~/deployments# kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
apache-5ffcd65847-6mcb5            1/1     Running   0           137m
apache-5ffcd65847-8tz6c            1/1     Running   0           106m
apache-5ffcd65847-htrbg            1/1     Running   0           137m
mariadb-5cdd94f94d-8xx9f          1/1     Running   0           106m
mariadb-5cdd94f94d-lfcdz          1/1     Running   0           114m
mariadb-5cdd94f94d-qtrcz          1/1     Running   0           114m
nginx-54bbbd955c-4hbg8             0/1     Pending   0           9s
nginx-54bbbd955c-ccpvc             0/1     Pending   0           9s
nginx-54bbbd955c-dv5jc             0/1     Pending   0           9s
root@kubes-01:~/deployments#

```

```

root@kubes-01:~/deployments# kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
apache-5ffcd65847-6mcb5            1/1     Running   0           137m
apache-5ffcd65847-8tz6c            1/1     Running   0           106m
apache-5ffcd65847-htrbg            1/1     Running   0           137m
mariadb-5cdd94f94d-8xx9f          1/1     Running   0           106m
mariadb-5cdd94f94d-lfcdz          1/1     Running   0           114m
mariadb-5cdd94f94d-qtrcz          1/1     Running   0           114m
nginx-54bbbd955c-4hbg8             0/1     Pending   0           9s
nginx-54bbbd955c-ccpvc             0/1     Pending   0           9s
nginx-54bbbd955c-dv5jc             0/1     Pending   0           9s
root@kubes-01:~/deployments# kubectl describe pvc nginx-pvc
Name:                               nginx-pvc
Namespace:                          default
StorageClass:                       local-path
Status:                              Bound
Volume:                             pvc-181233fa-9e0f-499a-af6e-9f7fd3daf79c

```

Mon PVC est Bound.

```

root@kubes-01:~/deployments# kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
apache-5ffcd65847-6mcb5            1/1     Running   0           140m
apache-5ffcd65847-8tz6c            1/1     Running   0           108m
apache-5ffcd65847-htrbg            1/1     Running   0           140m
mariadb-5cdd94f94d-8xx9f          1/1     Running   0           108m
mariadb-5cdd94f94d-lfcdz          1/1     Running   0           117m
mariadb-5cdd94f94d-qtrcz          1/1     Running   0           117m
nginx-54bbbd955c-4hbg8             1/1     Running   0           2m55s
nginx-54bbbd955c-ccpvc             1/1     Running   0           2m55s
nginx-54bbbd955c-dv5jc             1/1     Running   0           2m55s
root@kubes-01:~/deployments#

```

Tous mes Pods sont Running ! Maintenant, je vais faire les volumes persistants pour MariaDB.

```

root@kubes-01:~/deployments# kubectl get pods
NAME                                READY   STATUS    RESTARTS
AGE
apache-5ffcd65847-6mcb5             1/1     Running   0
149m
apache-5ffcd65847-8tz6c             1/1     Running   0
118m
apache-5ffcd65847-htrbg             1/1     Running   0
149m
mariadb-84745cd4b-2lcgc             1/1     Running   0
61s
mariadb-84745cd4b-5jq7f             1/1     Running   1 (17s ago)
61s
mariadb-84745cd4b-r9vvq             0/1     CrashLoopBackOff 1 (15s ago)
61s
nginx-54bbbd955c-4hbg8              1/1     Running   0
12m
nginx-54bbbd955c-ccpvc              1/1     Running   0
12m
nginx-54bbbd955c-dv5jc              1/1     Running   0
12m
root@kubes-01:~/deployments#

```

On peut voir qu'il y a un de mes pods qui a Crash. C'est normal, plusieurs pods ne peuvent pas écrire sur le même volume en même temps. Je vais donc changer à un seul volume.

Une fois le nombre de volumes changé, passant de 3 à 1, j'ai attendu que tout se stabilise et on peut vérifier que tout est bien en Running et que tout fonctionne.

```

root@kubes-01:~/deployments# kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
apache-5ffcd65847-6mcb5             1/1     Running   0           160m
apache-5ffcd65847-8tz6c             1/1     Running   0           129m
apache-5ffcd65847-htrbg             1/1     Running   0           160m
mariadb-84745cd4b-w7b6b             1/1     Running   0           3m10s
nginx-54bbbd955c-4hbg8              1/1     Running   0           23m
nginx-54bbbd955c-ccpvc              1/1     Running   0           23m
nginx-54bbbd955c-dv5jc              1/1     Running   0           23m
root@kubes-01:~/deployments#

```

Et on peut voir que tout est bien Bound.

```

root@kubes-01:~/deployments# kubectl get pvc
NAME                                STATUS    VOLUME                                     CAPACITY   ACCESS MODES   STORAGECLASS   VOLUMEATTRIBUTESCLASS   AGE
mariadb-pvc                        Bound     pvc-d58da178-2630-4b8c-b347-37ea4bbb5f5c   1Gi         RWO             local-path     <unset>                20m
nginx-pvc                          Bound     pvc-181233fa-9e0f-499a-af6e-9f7fd3daf79c   1Gi         RWO             local-path     <unset>                27m
root@kubes-01:~/deployments#

```

Job 6

Je vais créer une ConfigMap pour Nginx.

```
GNU nano 7.2 /root/deployments/nginx-configmap.yaml
apiVersion: v1
kind: ConfigMap
metadata:
  name: nginx-config
data:
  nginx.conf: |
    server {
      listen 8080;
      server_name localhost;
      location / {
        root /usr/share/nginx/html;
        index index.html index.htm;
      }
    }
  index.html: |
    <!DOCTYPE html>
    <html>
    <head><title>Kubernetes Project</title></head>
    <body>
      <h1>Bienvenue sur mon cluster Kubernetes !</h1>
      <p>Deploye avec K3S et ConfigMap</p>
    </body>
    </html>
```

```
root@kubes-01:~/deployments# kubectl get configmaps
NAME              DATA  AGE
kube-root-ca.crt  1      25h
nginx-config      2      4s
root@kubes-01:~/deployments#
```

```
root@kubes-01:~/deployments# kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
apache-5ffcd65847-6mcb5            1/1     Running   0           174m
apache-5ffcd65847-8tz6c            1/1     Running   0           143m
apache-5ffcd65847-htrbg            1/1     Running   0           174m
mariadb-84745cd4b-w7b6b            1/1     Running   0           16m
nginx-6574946749-474df             1/1     Running   0           4s
nginx-6574946749-ghtjq             1/1     Running   0           4s
nginx-6574946749-hh28t             1/1     Running   0           4s
root@kubes-01:~/deployments# curl http://192.168.88.150:30080
```

On voit bien que ça Running, mais par contre je ne peux pas accéder au site lorsque je fais mon curl en dessous. Je devrais voir un message, mais il n'y a rien. Le problème, c'est que le ConfigMap monte nginx et index.html, mais ne trouve pas de fichier config valide. Je vais donc séparer les deux,

```
k8s@kube x k8s@kube: X Windows F X + v - □ X
GNU nano 7.2 /root/deployments/nginx-configmap.yaml
apiVersion: v1
kind: ConfigMap
metadata:
  name: nginx-config
data:
  default.conf: |
    server {
      listen 8080;
      server_name localhost;
      location / {
        root /usr/share/nginx/html;
        index index.html index.htm;
      }
    }
}
```

Je vais maintenant créer un second ConfigMap pour le HTML.

```
root@kubes-01:~/deployments# kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
apache-5ffcd65847-6mcb5             1/1     Running   0           3h19m
apache-5ffcd65847-8tz6c             1/1     Running   0           168m
apache-5ffcd65847-htrbg             1/1     Running   0           3h19m
mariadb-84745cd4b-w7b6b             1/1     Running   0           41m
nginx-754cc87fc8-bxc6f              1/1     Running   0           10s
nginx-754cc87fc8-cwkmg              1/1     Running   0           10s
nginx-754cc87fc8-hk7q7              1/1     Running   0           10s
root@kubes-01:~/deployments# curl http://192.168.88.150:30080
<!DOCTYPE html>
<html>
<head><title>Kubernetes Project</title></head>
<body>
  <h1>Bienvenue sur mon cluster Kubernetes !</h1>
  <p>Deploye avec K3S et ConfigMap</p>
</body>
</html>
root@kubes-01:~/deployments# |
```

Et on peut voir que tout fonctionne parfaitement, et que ma page HTML s'affiche bel et bien.

Job 7

Je vais créer le secret MariaDB.

```
k8s@k8s: kube x k8s@k8s: kube x Windows F x + v - □ x
GNU nano 7.2 /root/deployments/mariadb-secret.yaml *
apiVersion: v1
kind: Secret
metadata:
  name: mariadb-secret
type: Opaque
stringData:
  root-password: "rootpassword"
  database: "mydb"
  user: "k8suser"
  password: "k8spassword"
```

```
root@kubes-01:~/deployments# kubectl get secrets
NAME                TYPE                      DATA  AGE
dockerhub-secret    kubernetes.io/dockerconfigjson  1      19h
mariadb-secret      Opaque                    4       3s
root@kubes-01:~/deployments#
```

Le secret est bien créé ! Je vais maintenant utiliser le secret dans le déploiement MariaDB.

```
root@kubes-01:~/deployments# kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
apache-5ffcd65847-6mcb5             1/1     Running   0           3h30m
apache-5ffcd65847-8tz6c             1/1     Running   0           179m
apache-5ffcd65847-htrbg             1/1     Running   0           3h30m
mariadb-6964f9cf7b-cbg98            1/1     Running   0           5s
nginx-754cc87fc8-bxc6f              1/1     Running   0           11m
nginx-754cc87fc8-cwkmg              1/1     Running   0           11m
nginx-754cc87fc8-hk7q7              1/1     Running   0           11m
root@kubes-01:~/deployments#
```

MariaDB tourne maintenant avec les credentials stockés dans un Secret.

```
root@kubes-01:~/deployments# kubectl describe pod mariadb-6964f9cf7b-cbg98 | grep -A 5 "Environment"
Environment:
  MYSQL_ROOT_PASSWORD: <set to the key 'root-password' in secret 'mariadb-secret'> Optional: false
  MYSQL_DATABASE: <set to the key 'database' in secret 'mariadb-secret'> Optional: false
  MYSQL_USER: <set to the key 'user' in secret 'mariadb-secret'> Optional: false
  MYSQL_PASSWORD: <set to the key 'password' in secret 'mariadb-secret'> Optional: false
Mounts:
root@kubes-01:~/deployments#
```

les variables d'environnement référencent le Secret sans afficher les valeurs en clair.

Job 8

Je vais créer un Service Account.

```

root@kubes-01:~/deployments# kubectl get serviceaccounts
NAME      AGE
default   26h
k8s-user  5s
root@kubes-01:~/deployments# kubectl get roles
NAME          CREATED AT
pod-reader    2026-06-09T11:57:02Z
root@kubes-01:~/deployments# kubectl get rolebindings
NAME                ROLE          AGE
pod-reader-binding  Role/pod-reader  16s
root@kubes-01:~/deployments#

```

On voit bien que ça a été créé ! J'ai fait :
mon ServiceAccount k8s-user, mon role pod-reader (peut lire les pods, services et deployments), et mon roleBinding pod-reader-binding (lie k8s-user au role pod-reader).

Je vérifie les permissions, je vais tester si la RBAC fonctionne correctement :

```

root@kubes-01:~/deployments# kubectl auth can-i get pods --as=system:serviceaccount:default:k8s-user
yes
root@kubes-01:~/deployments# kubectl auth can-i delete pods --as=system:serviceaccount:default:k8s-user
no
root@kubes-01:~/deployments# kubectl auth can-i get services --as=system:serviceaccount:default:k8s-user
yes
root@kubes-01:~/deployments# kubectl auth can-i create deployments --as=system:serviceaccount:default:k8s-user
no
root@kubes-01:~/deployments#

```

C'est exactement ce qu'on me demandait.

- ServiceAccount k8s-user créé
- Role pod-reader créé (lecture seule sur pods, services, deployments)
- RoleBinding créé (lie k8s-user au role)
- Permissions vérifiées avec kubectl auth can-i

Job 9

j'installe d'abord Helm sur kubes-01.

```

root@kubes-01:~/deployments# curl https://raw.githubusercontent.com/helm/helm/main/scripts/get-helm-3 | bash
% Total    % Received % Xferd  Average Speed   Time    Time     Time  T
ime Current                        Dload  Upload  Total  Spent  L
eft  Speed
  0     0    0     0    0     0      0      0 --:--:-- --:--:-- --:
100 11929 100 11929    0     0 49452      0 --:--:-- --:--:-- --:
--:-- 52783
[WARNING] Could not find git. It is required for plugin installation.
Downloading https://get.helm.sh/helm-v3.21.0-linux-amd64.tar.gz
Verifying checksum... Done.
Preparing to install helm into /usr/local/bin
helm installed into /usr/local/bin/helm
root@kubes-01:~/deployments# helm version
version.BuildInfo{Version:"v3.21.0", GitCommit:"e0878d41b711792be60777fd65ad23a101e6b85f", GitTreeState:"clean", GoVersion:"go1.25.10"}
root@kubes-01:~/deployments#

```

J'ajoute ensuite un Repository Helm.

```

root@kubes-01:~/deployments# helm repo add bitnami https://charts.bitnami.com/bitnami
"bitnami" has been added to your repositories
root@kubes-01:~/deployments# helm repo update
Hang tight while we grab the latest from your chart repositories...
...Successfully got an update from the "bitnami" chart repository
Update Complete. ✨Happy Helming!✨
root@kubes-01:~/deployments# helm repo list
NAME      URL
bitnami   https://charts.bitnami.com/bitnami
root@kubes-01:~/deployments#

```

```

root@kubes-01:~/deployments# helm install mon-nginx bitnami/nginx --set service.type=NodePort --set service.nodePorts.http=30090
Error: INSTALLATION FAILED: Kubernetes cluster unreachable: Get "http://localhost:8080/version": dial tcp [::1]:8080: connect: connection refused
root@kubes-01:~/deployments#

```

J'ai eu un problème de configuration kubectl pour Helm, je vais donc spécifier mon fichier de configuration.


```

root@kubes-01:/home/k8s# kubectl get pods
NAME                                READY   STATUS    RESTARTS   A
GE
apache-5ffcd65847-6mcb5             1/1     Running   1 (7m49s ago)   1
0h
apache-5ffcd65847-8tz6c             1/1     Running   1 (7m49s ago)   1
0h
apache-5ffcd65847-htrbg             1/1     Running   1 (8m41s ago)   1
0h
mariadb-6964f9cf7b-cbg98            1/1     Running   1 (7m49s ago)   7
h9m
mon-nginx-7b5cdf4f4d-sgcnc          1/1     Running   0               8
9s
nginx-754cc87fc8-bxc6f              1/1     Running   1 (8m41s ago)   7
h20m
nginx-754cc87fc8-cwkmg              1/1     Running   1 (7m49s ago)   7
h20m
nginx-754cc87fc8-hk7q7              1/1     Running   1 (7m50s ago)   7
h20m
root@kubes-01:/home/k8s# helm list
Error: Kubernetes cluster unreachable: Get "http://localhost:8080/version": dial tcp [::1]:8080: connect: connection refused
root@kubes-01:/home/k8s#

```

Le pod est running maintenant ! Pour helm list, je vais re-exporter la variable.

```

root@kubes-01:/home/k8s# helm list
Error: Kubernetes cluster unreachable: Get "http://localhost:8080/version": dial tcp [::1]:8080: connect: connection refused
root@kubes-01:/home/k8s# export KUBECONFIG=/etc/rancher/k3s/k3s.yaml
root@kubes-01:/home/k8s# helm list
NAME                                NAMESPACE    REVISION    UPDATED          CHART      A
PP VERSION
mon-nginx                          default      1           2026-06-09 20:57  nginx-25.0.2  1
:53.437716642 +0200 CEST          deployed
.31.1
root@kubes-01:/home/k8s# echo 'export KUBECONFIG=/etc/rancher/k3s/k3s.yaml' >> /root/.bashrc
root@kubes-01:/home/k8s# source /root/.bashrc
root@kubes-01:/home/k8s# |

```

Maintenant, helm fonctionne ! Je vais vérifier que nginx Helm réponde bien :

```
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, nginx is successfully installed and work
ing.
Further configuration is required for the web server, reverse pro
xy,
API gateway, load balancer, content cache, or other features.</p>

<p>For online documentation and support please refer to
<a href="https://nginx.org/">nginx.org</a>.<br/>
To engage with the community please visit
<a href="https://community.nginx.org/">community.nginx.org</a>.<b
r/>
For enterprise grade support, professional services, additional
security features and capabilities please refer to
<a href="https://f5.com/nginx">f5.com/nginx</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
root@kubes-01:/home/k8s#
```

Et tout fonctionne ! Je peux maintenant le personnaliser avec des valeurs.

```
root@kubes-01:/home/k8s# helm list
NAME                NAMESPACE      REVISION    UPDATED             CHART          APP VERSI
ON
mon-nginx           default         2           2026-06-09 21:04:34.448914471 +0200 CEST deployed        nginx-25.0.2   1.31.1
```

Helm a bien mis à jour le déploiement avec deux replicas !

```
root@kubes-01:/home/k8s# kubectl get pods
NAME                READY    STATUS    RESTARTS    AGE
apache-5ffcd65847-6mcb5    1/1     Running    1 (14m ago)    10h
apache-5ffcd65847-8tz6c    1/1     Running    1 (14m ago)    10h
apache-5ffcd65847-htrbg    1/1     Running    1 (15m ago)    10h
mariadb-6964f9cf7b-cbg98   1/1     Running    1 (14m ago)    7h15m
mon-nginx-7b5cdf4f4d-4cfs7  1/1     Running    0             77s
mon-nginx-7b5cdf4f4d-sgcnc  1/1     Running    0             7m57s
nginx-754cc87fc8-bxc6f     1/1     Running    1 (15m ago)    7h27m
nginx-754cc87fc8-cwkmg     1/1     Running    1 (14m ago)    7h27m
nginx-754cc87fc8-hk7q7     1/1     Running    1 (14m ago)    7h27m
root@kubes-01:/home/k8s#
```

Les deux pods mon-nginx sont “Running” maintenant ! On va pouvoir passer à la désinstallation.

```
root@kubes-01:/home/k8s# helm uninstall mon-nginx
release "mon-nginx" uninstalled
root@kubes-01:/home/k8s# kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
apache-5ffcd65847-6mcb5             1/1     Running   1 (15m ago)  10h
apache-5ffcd65847-8tz6c             1/1     Running   1 (15m ago)  10h
apache-5ffcd65847-htrbg             1/1     Running   1 (16m ago)  10h
mariadb-6964f9cf7b-cbg98           1/1     Running   1 (15m ago)  7h16m
nginx-754cc87fc8-bxc6f             1/1     Running   1 (16m ago)  7h28m
nginx-754cc87fc8-cwkmg             1/1     Running   1 (15m ago)  7h28m
nginx-754cc87fc8-hk7q7             1/1     Running   1 (15m ago)  7h28m
root@kubes-01:/home/k8s#
```

Et voilà, j'ai tout fini, et j'ai bien terminé le projet.